Technical Summary:
LUQ: Long-text Uncertainty Quantification for LLMs

Kai-Yu Lu

1 Research Problem and Motivation

1.1 Problem Setting

Long-text generation by Large Language Models (LLMs) often contains *nonfactual content* (hallucination). Uncertainty Quantification (UQ) aims to estimate how confident a model is about its generated output, so that unreliable generations can be detected and potentially filtered. Existing UQ research in LLMs has largely focused on *short* generations, while many real-world applications require *long-form* passages.

1.2 Why Existing Methods Are Insufficient

Two gaps motivate the work:

- **Length mismatch:** Many UQ approaches are evaluated on short, word-limited outputs. For long passages, small local errors can appear sparsely, and surface-level consistency can degrade due to paraphrasing and stylistic variation.
- **Access constraints:** A substantial portion of prior UQ methods require access to internal model states such as logits (White-Box UQ). This assumption fails for closed-source LLMs, motivating Black-Box UQ methods that rely only on sampled outputs. Existing Black-Box methods are not designed to robustly handle long-form content.

1.3 Task Definition and Success Criterion

The target is **output uncertainty** (uncertainty induced by the generated text itself). Success is defined as an uncertainty score that is **negatively correlated** with the factuality of the generated answer. Formally, for two queries x_i and x_j:

$$U(x_i) \leq U(x_j) \iff F(x_i) \geq F(x_j), \tag{1}$$

where $U(\cdot)$ is the uncertainty score and $F(\cdot)$ is a factuality score.

Purpose (Eq. 1). Defines the desired monotonic relationship between uncertainty and factuality for evaluation.

Symbols (Eq. 1). x_i, x_j: user queries; $U(x)$: uncertainty score; $F(x)$: factuality score.

Intuition (Eq. 1). More factual answers should be assigned lower uncertainty.

Usage (Eq. 1). Operationalized via correlation (Pearson and Spearman) between U and F across a benchmark.

2 Related Work

2.1 Uncertainty Quantification for Natural Language Generation

UQ in Natural Language Generation (NLG) broadly includes White-Box approaches (requiring logits or internal activations) and Black-Box approaches (using only generated texts). White-Box methods are not applicable to closed-source LLMs. Black-Box methods often estimate uncertainty through disagreement among multiple sampled generations, but are typically designed and validated on short responses.

2.2 Factuality Evaluation for Long-form Generation

Long-form factuality evaluation is challenging. Traditional overlap-based metrics are insufficient for factuality in free-form text. The study adopts FACTSCORE, an automatic factuality evaluation framework designed for long-form responses, and additionally validates it with a small human annotation study.

2.3 Baseline Families Used in This Study

Table 1 summarizes the UQ baselines evaluated.

3 Dataset Construction

3.1 Primary Benchmark: FACTSCORE-BIO

FACTSCORE-BIO contains 500 biography queries, where each query asks for a biography of an entity drawn from Wikidata with a corresponding Wikipedia entry. Wikipedia is used as the primary knowledge source for evaluation.

3.2 New Benchmark: FACTSCORE-DIS

To test domain transfer, a new dataset FACTSCORE-DIS is constructed for disease entities, following the methodology of the original FACTSCORE dataset. Disease names are collected via a Wikidata query, and entities with empty Wikipedia pages are removed. The paper provides construction details, including frequency category definitions, but does not specify the final dataset size in the main text.

3.3 Dataset Summary

4 Query Protocol and Task Definitions

4.1 Inputs, Outputs, and Sampling Protocol

For each query x, the system obtains one primary response r_a (treated as the model answer to be evaluated for factuality), and additional stochastic sample responses $R' = \{r_1, \ldots, r_n\}$ for uncertainty estimation. The number of stochastic samples n is treated as a tunable parameter, and the paper analyzes the impact of varying the number of samples.

Table 1: Baseline UQ methods and access assumptions. Items marked as White-Box require logits and are not applicable to closed-source LLMs.

Family	Methods (Full Term and ABBR)	Access
White-Box	Maximum Softmax Probability (MSP); Monte Carlo Sequence Entropy (MCSE); Semantic Entropy (SE)	Logits
Black-Box	Lexical Similarity (LexSim); Embedding Clustering Coefficient (Ecc); Number of Semantic Sets (NumSets); Eigenvalue-based score (EigV); Degree-based score (Deg); SelfCheckNLI (SCN)	Samples only
Proposed	Long-text Uncertainty Quantification (LUQ); LUQ-ATOMIC; LUQ-PAIR; LUQ-ENSEMBLE	Samples only (LUQ-ATOMIC uses an external atomic-fact splitter)

4.2 Factuality and Uncertainty Metrics

- **Factuality Score (FS):** FACTSCORE computes a score for the factuality of r_a.
- **Penalized Factuality Score (PFS):** To handle model refusals, refusals are assigned factuality 0.
- **Penalized Uncertainty Score (PUS):** Refusals are assigned uncertainty 1.
- **Correlation Metrics:** Pearson Correlation Coefficient (PCC) and Spearman Correlation Coefficient (SCC) between factuality (FS or PFS) and uncertainty (U or PUS) are reported. Strong negative correlation indicates effective uncertainty estimation.

4.3 Selective Question Answering Protocol

Selective question answering rejects answers with the highest uncertainty. The protocol abstains on a fixed percentile of queries based on the uncertainty score, and reports the resulting PFS and refusal rate.

Table 2: Datasets used in the study. Missing fields are reported as "Not specified in the paper."

Field	**FACTSCORE-BIO / FACTSCORE-DIS**
Name	FACTSCORE-BIO; FACTSCORE-DIS
Domain	Biographies (BIO); Diseases (DIS)
Source Entities	Wikidata with Wikipedia entries (BIO); Wikidata disease entities with Wikipedia pages (DIS)
Size	500 queries (BIO); Not specified in the paper (DIS)
Query Template	"Tell me a bio of {Person}." (BIO); Not specified in the paper (DIS query wording)
Knowledge Source for Evaluation	Wikipedia
Answer Type	Long-form passages
Preprocessing	Not specified in the paper (BIO); remove empty Wikipedia pages (DIS)
Splits	Not specified in the paper
Frequency Metadata	Wikipedia frequency categories used and analyzed (BIO); frequency category definitions provided (DIS)

5 Modeling Approach

5.1 Overview

LUQ is a sampling-based, Black-Box UQ method designed for long-form generation. It quantifies uncertainty by measuring **semantic inconsistency** across multiple sampled responses using Natural Language Inference (NLI). Natural Language Inference (NLI) is treated as a binary decision about whether one text entails another.

5.2 Step-by-step Pipeline

1. **Sampling:** Generate a set of stochastic responses $R' = \{r_1, \ldots, r_n\}$ for the same query x.
2. **Sentence Segmentation:** Split each response r_i into sentences $\{s_{i,1}, \ldots, s_{i,m_i}\}$.
3. **NLI-based Consistency Scoring:** For each sentence $s_{i,j}$, compute its entailment probability against a reference response r' (another sampled response). Aggregate sentence-level entailment to obtain a response-to-response similarity $S(r_i, r')$.
4. **Confidence and Uncertainty Aggregation:** Compute a confidence score for each sampled response from its similarity to other samples, then aggregate to a query-level uncertainty $U(x)$.

5.3 Core Formulas

$$P(\text{entail} \mid s, r') = \frac{\exp(l_e)}{\exp(l_e) + \exp(l_c)}. \tag{2}$$

Purpose (Eq. 2). Converts NLI model outputs into a scalar entailment probability used for semantic consistency scoring.

Symbols (Eq. 2). s: a sentence (hypothesis); r': a reference response (premise); l_e: entailment logit; l_c: contradiction logit; $P(\text{entail} \mid s, r') \in [0, 1]$.

Intuition (Eq. 2). If a sentence is supported by the reference response, entailment should dominate contradiction, producing a high probability.

Usage (Eq. 2). Computed for sentence-to-response comparisons inside LUQ; these values are averaged to form response-level similarity and then aggregated to uncertainty.

$$C(x, r_i) = \frac{1}{n} \sum_{r' \in R' \setminus \{r_i\}} S(r_i, r'), \qquad U(x) = \frac{1}{n+1} \sum_{r_i \in R'} (1 - C(x, r_i)) \,. \tag{3}$$

Purpose (Eq. 3). Defines query-level uncertainty from cross-sample semantic consistency.

Symbols (Eq. 3). x: query; $R' = \{r_1, \ldots, r_n\}$: sampled responses; r_i: one sample; $S(r_i, r')$: similarity between responses based on averaged entailment of sentences; $C(x, r_i)$: confidence for sample r_i; $U(x) \in [0, 1]$: uncertainty score.

Intuition (Eq. 3). If sampled answers agree semantically, each r_i is well supported by other samples, increasing C and reducing U. Disagreement across samples decreases C and increases U, signaling uncertainty.

Usage (Eq. 3). $U(x)$ is computed after sampling and NLI scoring, then used for correlation evaluation, selective answering, and ensemble selection.

$$P(\text{entail} \mid s_j, r') = \max_{s'_j \in r'} \left| P(\text{entail} \mid s_j, s'_j) \right| . \tag{4}$$

Purpose (Eq. 4). Implements LUQ-PAIR to mitigate NLI degradation when premises are long by matching each sentence to the most supportive sentence in the reference.

Symbols (Eq. 4). s_j: a sentence in the current response; s'_j: a sentence in reference response r'; $P(\text{entail} \mid s_j, s'_j)$: entailment probability for a sentence pair.

Intuition (Eq. 4). A long reference response can contain many sentences, and only one sentence needs to support s_j. Taking the maximum captures the best local support signal.

Usage (Eq. 4). Replaces sentence-to-response entailment in the similarity computation $S(\cdot,\cdot)$ within LUQ-PAIR.

5.4 Model Components and Training

The NLI model is DeBERTa-v3-large, fine-tuned on MultiNLI to better match the hypothesis (short sentence) and premise (longer text) format used in LUQ. LUQ-ATOMIC first uses ChatGPT to decompose a response into atomic fact units, then applies the same consistency computation at the atomic-fact level rather than sentence level.

5.5 Minimal Illustrative Example

Consider a query asking for a biography. Multiple stochastic samples can disagree on a specific fact such as a date or affiliation while remaining fluent and similar in style. LUQ splits each sample into sentences and checks whether each sentence is supported (entailed) by other samples. If several samples disagree on that specific sentence, the entailment scores drop, reducing cross-sample confidence and increasing the query uncertainty. If all samples consistently support the same set of sentences, the uncertainty remains low.

6 Empirical Results

6.1 Experimental Setup

Models. Six LLMs are evaluated: GPT-4, GPT-3.5, Gemini 1.0 Pro, Yi-34B-Chat, Tulu-2-70B, and Vicuna-33B.
Sampling. Responses are generated with temperature 0.7.
Infrastructure. Experiments are run on a machine with an NVIDIA A100 (80GB) and Intel Xeon Silver 4208 CPU.
Cost Discussion. The paper provides an estimate indicating that applying FACTSCORE with multiple samples can be expensive, with an example estimate around \$1000 for a dataset of 500 QA pairs under a specified sampling and verification setting.

6.2 Main UQ Quality: Correlation with Factuality

Table 3 reports PCC and SCC between uncertainty scores and factuality on FACTSCORE-BIO. LUQ achieves consistently strong negative correlations and outperforms Black-Box baselines across models. The abstract highlights a negative coefficient of approximately -0.85 for Gemini 1.0 Pro, indicating that high-factuality outputs tend to have low LUQ uncertainty.

Interpretation.

- **Strength:** LUQ improves correlation consistently, with Gemini 1.0 Pro reaching PCC -85.1 and SCC -78.7, indicating strong alignment between uncertainty and factuality.
- **Weakness:** White-Box methods are unavailable for closed-source models and show near-zero or positive correlations for open-source models in this setting, demonstrating limited effectiveness on long-form outputs under the evaluated protocol.

Table 3: FACTSCORE-BIO: Correlation between factuality and uncertainty (Table values reproduced from the paper). Higher magnitude negative correlation indicates better alignment with factuality. Dashes indicate methods not applicable to closed-source models.

	White-Box			Black-Box						
Model	MSP	MCSE	SE	LexSim	Ecc	NumSets	EigV	Deg	SCN	LUQ
PCC on FACTSCORE-BIO										
GPT-4	–	–	–	-45.2	-24.8	-8.24	-36.9	-3.78	-53.1	-60.4
GPT-3.5	–	–	–	-67.8	-10.6	-11.9	-30.3	-22.4	-65.1	-71.3
Gemini 1.0 Pro	–	–	–	-67.2	-50.3	-53.0	-72.7	-64.8	-78.7	-85.1
Yi-34B-Chat	2.96	2.49	-40.6	-50.7	-32.8	-35.5	-56.3	-40.9	-66.4	-73.8
Tulu-2-70B	4.95	4.39	-35.1	-67.9	-34.7	-33.5	-47.5	-51.0	-68.1	-77.6
Vicuna-33B	2.35	4.44	-33.4	-56.6	-43.2	-34.9	-43.1	-45.6	-68.5	-71.8
SCC on FACTSCORE-BIO										
GPT-4	–	–	–	-36.0	-12.7	4.18	-18.7	6.73	-41.8	-45.3
GPT-3.5	–	–	–	-52.4	-26.5	-17.0	-34.6	-22.9	-61.1	-66.6
Gemini 1.0 Pro	–	–	–	-61.8	-43.4	-49.3	-69.9	-62.1	-73.1	-78.7
Yi-34B-Chat	2.07	0.18	-41.1	-41.7	-21.4	-24.3	-46.7	-29.4	-64.0	-65.7
Tulu-2-70B	-0.50	1.74	-34.6	-57.1	-18.0	-17.9	-39.4	-41.6	-68.9	-74.4
Vicuna-33B	3.53	3.47	-34.6	-46.5	-37.1	-32.4	-40.0	-40.9	-66.7	-67.7

6.3 Domain Transfer: FACTSCORE-DIS

The study evaluates GPT-3.5 and Yi-34B-Chat on FACTSCORE-DIS and reports improved correlations for LUQ over baselines.

Table 4: FACTSCORE-DIS: Correlation between factuality and uncertainty (Table values reproduced from the paper).

	White-Box			Black-Box						
Model	MSP	MCSE	SE	LexSim	Ecc	NumSets	EigV	Deg	SCN	LUQ
PCC on FACTSCORE-DIS										
GPT-3.5	–	–	–	-36.0	-10.1	-19.5	-28.3	-25.7	-37.7	-54.5
Yi-34B-Chat	-13.2	-11.3	-44.2	-53.5	-37.8	-30.1	-60.1	-44.5	-71.4	-73.3
SCC on FACTSCORE-DIS										
GPT-3.5	–	–	–	-29.1	-6.55	-5.45	-18.6	-23.6	-36.5	-46.2
Yi-34B-Chat	-8.00	-5.64	-44.2	-43.5	-19.9	-15.2	-49.9	-34.4	-70.7	-72.2

6.4 Selective Question Answering

Table 5 reproduces selective answering results. Rejecting the top 15% most uncertain queries increases PFS for several models by more than 5 points, while increasing refusal rate by design.

Interpretation.

Table 5: Selective question answering with LUQ on FACTSCORE-BIO (values reproduced from the paper). "Percentile" indicates the fraction of most-uncertain queries abstained.

Model	**5th**		**10th**		**15th**	
	PFS	RR(%)	PFS	RR(%)	PFS	RR(%)
GPT-4	0.711	24.6	0.718	27.6	0.721	30.0
GPT-3.5	0.574	11.6	0.609	15.4	0.640	19.4
Gemini 1.0 Pro	0.470	18.0	0.534	25.8	0.584	32.2
Yi-34B-Chat	0.610	15.0	0.641	19.6	0.673	24.8
Tulu-2-70B	0.689	1.60	0.708	2.40	0.723	3.00
Vicuna-33B	0.475	3.80	0.519	5.00	0.557	6.20

- **Advantage:** LUQ enables a practical abstention policy that yields higher average factuality among answered queries.
- **Trade-off:** Factuality gains are coupled with increased refusal rates, reflecting a controllable quality-coverage trade-off rather than unconditional improvement.

6.5 LUQ-ENSEMBLE for Factuality Improvement

LUQ-ENSEMBLE ensembles responses from multiple LLMs and selects the response with the lowest uncertainty. Table 6 reproduces the reported results.

Table 6: LUQ-ENSEMBLE results (values reproduced from the paper). PFS: penalized factuality score. PUS: penalized uncertainty score. AD: answerability degree (percentage of queries answered).

Method and Candidate Set	**PFS**	**PUS**	**AD(%)**
Best LLM (Tulu-2-70B)	0.688	0.495	100
LUQ-ENSEMBLE (GPT-3.5, Yi-34B-Chat)	0.528	0.458	100
LUQ-ENSEMBLE (GPT-3.5, Gemini 1.0 Pro, Yi-34B-Chat)	0.588	0.376	100
LUQ-ENSEMBLE (GPT-4, Tulu-2-70B)	0.794	0.271	100
LUQ-ENSEMBLE (GPT-4, Gemini 1.0 Pro, Tulu-2-70B)	0.826	0.236	100

Interpretation.

- **Strength:** Adding LUQ-based selection on top of strong LLMs increases PFS, with the best reported ensemble (GPT-4, Gemini 1.0 Pro, Tulu-2-70B) achieving PFS 0.826 compared to 0.688 for the best standalone LLM (Tulu-2-70B).
- **Constraint:** LUQ-ENSEMBLE requires running multiple LLMs per query, increasing compute and latency.

6.6 Ablations and Robustness

Two LUQ variations improve correlation at increased cost:

- **LUQ-PAIR:** Improves correlation by matching each sentence to the best supporting sentence in the reference, but increases NLI computations from $M \times M$ to $N \times M^2$ for N samples and M sentences.
- **LUQ-ATOMIC:** Improves correlation further by using atomic facts. On FACTSCORE-BIO, Pearson correlation improves across models, for example from -85.1 (LUQ) to -87.2 (LUQ-ATOMIC) for Gemini 1.0 Pro.

6.7 Human Validation of FACTSCORE

A small human evaluation reports Fleiss' Kappa 0.793 for inter-annotator agreement, and Pearson correlation 0.88 between FACTSCORE and human factuality judgments on 50 sampled passages, supporting FACTSCORE as an evaluation signal in these experiments.

7 Summary

7.1 Contributions and Innovations Compared to Prior Work

- **Long-text specific UQ design:** LUQ introduces a sentence-level NLI-based consistency framework tailored to long passages, addressing weaknesses of similarity-based disagreement measures under paraphrase and length.
- **Black-Box applicability:** LUQ operates without logits, enabling evaluation on closed-source LLMs where White-Box baselines are inapplicable.
- **Domain extension:** FACTSCORE-DIS extends evaluation beyond biographies into a medical domain setting using a dataset constructed following FACTSCORE methodology.
- **Actionable usage protocols:** Selective answering and LUQ-ENSEMBLE demonstrate how uncertainty estimates can translate into improved factuality through abstention or cross-model selection.

7.2 Limitations and Future Directions (As Stated or Implied by Reported Experiments)

- **Computational cost:** LUQ requires multiple sampled responses and many NLI computations; LUQ-PAIR further increases complexity, and LUQ-ATOMIC introduces additional external calls.
- **Evaluation fairness concern for LUQ-ATOMIC:** LUQ-ATOMIC and FACTSCORE both use ChatGPT-based atomic decomposition, which can create an unfair comparison against baselines that do not use this step.
- **Controlled uncertainty scope:** The study focuses on output uncertainty under relatively clear, answerable questions, and does not specify broader coverage of input uncertainty or inherently ambiguous QA benchmarks.
- **FACTSCORE dependence:** Results rely on FACTSCORE as the factuality reference, with a supplementary but limited human validation.